

9th Set of Problems

1. Consider the robotic arm in Figure 1. Following the Newton-Euler methodology, determine the equations of the motion of the manipulator. Neglect the mass and inertia of the motors and all friction forces.

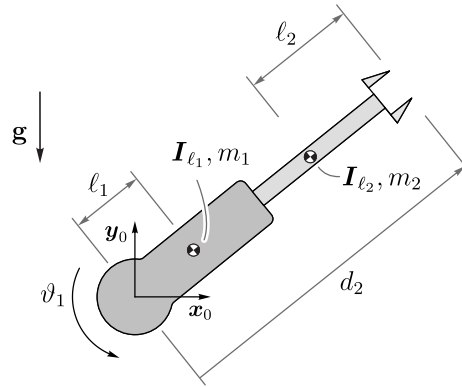


Figure 1: Polar manipulator.

2. Consider the manipulator in Figure 2(a). Following the Newton-Euler methodology, determine the equations of the motion of the manipulator. Consider the mass and inertia of each link, and neglect the mass and inertia of the motors and any friction forces.

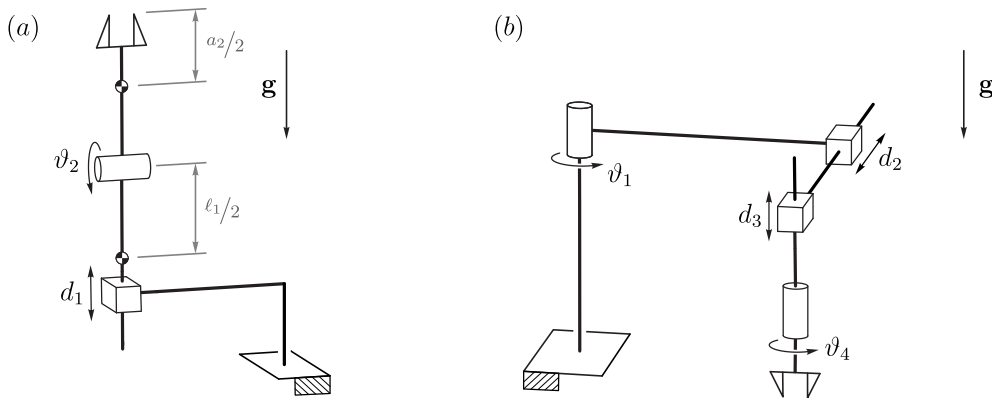


Figure 2: (a) Stadium manipulator. (b) Cylindrical manipulator.

3. Consider the robotic arm in Figure 2(b), with only the first two joints and links. Neglect the mass and inertia of the motors and all friction forces. Consider the mass and inertia of each link, and assume the link's center of mass is located at their center. Following the Newton-Euler methodology, determine the equations of the motion of the manipulator.